

<i>Title:</i>	Revision of the Bachelor of Science in Chemical Engineering
<i>Sponsor:</i>	<i>Department of Chemical Engineering</i>
<i>Executive Summary:</i>	Additional course are included as options to CHEM 222 Analytical Chemistry in the Nonengineering and General Education degree requirements for Honors College students in the BS in Chemical Engineering program to take into account the revision to the Honors College General Chemistry sequence (CHEM116-118)
<i>Description:</i>	<i>Add a footnote to CHEM 222 Analytical Chemistry describing the course options that Honors College students can take instead of CHEM 222 to fulfill the Nonengineering and General Education Requirements:</i> <i>CHEM 222 Analytical Chemistry^c</i> <i>c Honors College students taking CHEM116 and CHEM118 can select CHEM 222 or select one option from the following courses:</i> <i>CHEM 314 Inorganic Chemistry (4 hours)</i> <i>CHEM 452 Biochemistry (4 hours)</i> <i>CHEM 402 Chemical Information Systems (2 hrs) + CHEM 444 Physical Chemistry III (2 hrs)</i>
<i>Justification:</i>	<i>Chemistry 116-118 are being revised to include content covered in analytical chemistry. As a result, students who successfully complete both Chemistry 116 and 118 will have fulfilled any requirement for general chemistry and for analytical chemistry (CHEM 222). This only affects chemical engineering Honors College students who must take CHEM116/118 instead of CHEM 112/114. Since they do not need to take CHEM 222 Analytical Chemistry, this results in a decrease of 4 credit hours for the Nonengineering and General Education Requirement. In order to fulfill the requirements, Honors College students are given the option to either take CHEM 222 (extra analytical practice) or the option of other Chemistry courses (CHEM 314, CHEM 452, or CHEM 402+CHEM444).</i>
<i>Catalog Statement:</i>	<i>See below</i>
<i>Minority Impact Statement:</i>	<i>N/A</i>

<i>Budgetary and Staff Implications:</i>	N/A
<i>Library Resource Implications:</i>	N/A
<i>Space Implications:</i>	N/A
<i>Unit (e.g. department) approval date:</i> <i>College (educational policy committee, faculty) approval dates:</i>	
<i>Contact Person:</i>	Alan Zdunek zdunek@uic.edu
<i>Proposed Effective Date/Term:</i>	<i>Example:</i> Fall, 2016

Current	Revised										
BS in Chemical Engineering <i>Degree Requirements</i> To earn a Bachelor of Science in Chemical Engineering degree from UIC, students need to complete University, college, and department degree requirements. The Department of Chemical Engineering degree requirements are outlined below. Students should consult the <i>College of Engineering</i> section for additional degree requirements and college academic policies. BS in Chemical Engineering Degree Requirements Hours <table> <tr> <td>Nonengineering and General Education Requirements</td><td>74</td></tr> <tr> <td>Required in the College of Engineering</td><td>48</td></tr> <tr> <td>Technical Elective^a</td><td>3</td></tr> <tr> <td>Electives outside the Major Rubric^a</td><td>3</td></tr> <tr> <td>Total Hours—BS in Chemical Engineering</td><td>128</td></tr> </table> ^a <i>Students in the Biochemical Engineering Concentration take a minimum of 8 hours of electives and 130 hours for the degree; see below.</i>	Nonengineering and General Education Requirements	74	Required in the College of Engineering	48	Technical Elective ^a	3	Electives outside the Major Rubric ^a	3	Total Hours—BS in Chemical Engineering	128	BS in Chemical Engineering SAME SAME
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Nonengineering and General Education Requirements <table border="1"> <thead> <tr> <th>Courses</th><th>Hours</th></tr> </thead> <tbody> <tr><td>ENGL 160— Academic Writing I: Writing for Academic and Public Contexts</td><td>3</td></tr> <tr><td>ENGL 161— Academic Writing II: Writing for Inquiry and Research</td><td>3</td></tr> <tr><td>Exploring World Cultures course^a</td><td>3</td></tr> <tr><td>Understanding the Creative Arts course^a</td><td>3</td></tr> <tr><td>Understanding the Past course^a</td><td>3</td></tr> <tr><td>Understanding the Individual and Society course^a</td><td>3</td></tr> <tr><td>Understanding U.S. Society course^a</td><td>3</td></tr> <tr><td>MATH 180—Calculus I^b</td><td>5</td></tr> <tr><td>MATH 181—Calculus II^b</td><td>5</td></tr> <tr><td>MATH 210—Calculus III^b</td><td>3</td></tr> <tr><td>MATH 220—Introduction to Differential Equations</td><td>3</td></tr> <tr><td>PHYS 141—General Physics I (Mechanics)^b</td><td>4</td></tr> <tr><td>PHYS 142—General Physics II (Electricity and Magnetism)^b</td><td>4</td></tr> <tr><td>CHEM 112—General College Chemistry I^b</td><td>5</td></tr> <tr><td>CHEM 114—General College Chemistry II^b</td><td>5</td></tr> <tr><td>CHEM 222—Analytical Chemistry</td><td>4</td></tr> <tr><td>CHEM 232—Organic Chemistry I</td><td>4</td></tr> <tr><td>CHEM 233—Organic Chemistry Laboratory I</td><td>1</td></tr> <tr><td>CHEM 234—Organic Chemistry II</td><td>4</td></tr> <tr><td>CHEM 342—Physical Chemistry I</td><td>3</td></tr> <tr><td>CHEM 346—Physical Chemistry II</td><td>3</td></tr> <tr><td>Total Hours—Nonengineering and General Education Requirements</td><td>74</td></tr> </tbody> </table> a Students should consult the General Education section of the catalog for a list of approved courses in this category. b This course is approved for the Analyzing the Natural World General Education category. 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BS in Chemical Engineering, Bioengineering Concentration Students in this concentration complete the following:			SAME
Required Courses ^a			
Technical Elective			
<u>CHE 422</u>	Biochemical Engineering	3	
Electives			
Select two electives in nonmajor rubric category from the following:		5-7	
<u>BIOS 350</u>	General Microbiology		
<u>BIOS 351</u>	Microbiology Laboratory		
<u>CHEM 352</u>	Introductory Biochemistry		
<u>CHEM 452</u>	Biochemistry I		
Total Hours		8-10	
^a Due to structure of the concentration and the prerequisites required for some of the courses, students in the concentration will be required to take a minimum of 130 semester hours for the degree.			
Sample Course Schedule			
Freshman Year			
First Semester		Hours	SAME
MATH 180—Calculus I		5	
CHEM 112—General College Chemistry I		5	
ENGL 160—Academic Writing I: Writing for Academic and Public Contexts		3	
CS 109—C/C++ Programming for Engineers with MatLab		3	
ENGR 100—Orientation ^a		1 ^a	
Total Hours		16	
Second Semester		Hours	SAME
MATH 181—Calculus II		5	
PHYS 141—General Physics I (Mechanics)		4	
ENGL 161—Academic Writing II: Writing for Inquiry and Research		3	
CHEM 114—General College Chemistry II		5	
Total Hours		17	

Sophomore Year		
First Semester	Hours	SAME
MATH 210—Calculus III	3	
PHYS 142—General Physics II (Electricity and Magnetism)	4	
CHEM 232—Organic Chemistry I	4	
CHE 201—Introduction to Thermodynamics	3	
General Education Core course	3	
Total Hours	17	
Second Semester	Hours	SAME
MATH 220—Introduction to Differential Equations	3	
CHEM 234—Organic Chemistry II	4	
CHEM 233—Organic Chemistry Lab I	1	
CHE 205—Computational Methods in Chem. Eng.	3	
CHE 210—Material and Energy Balances	4	
CME 260— Properties of Materials	3	
Total Hours	18	
Junior Year		
First Semester	Hours	SAME
ECE 210—Electrical Circuit Analysis	3	
CHE 301—ChE Thermodynamics	3	
CHE 311—Transport Phenomena I	3	
CHEM 222—Analytical Chemistry	4	
General Education Core course	3	
Total Hours	16	
Second Semester	Hours	SAME
CHEM 342—Physical Chemistry I	3	
CHE 312—Transport Phenomena II	3	
CHE 313— Transport Phenomena III	3	
CHE 321—Chemical Reaction Engineering	3	
General Education Core course	3	
Total Hours	15	
Senior Year		
First Semester	Hours	SAME
CHE 381—Chemical Engineering Laboratory I	2	
CHE 396—Senior Design I	4	
CHEM 346—Physical Chemistry II	3	
CHE Technical elective—Selected from CHE 410, 413, 421, 422, 423, 431, 438, 440, 441, 445, 450, 456, 494, or 392 (departmental approval is required for CHE 392)	3	
General Education Core Course	3	
Total Hours	15	
Second Semester	Hours	
CHE 382—Chemical Engineering Laboratory II	2	
CHE 341—Chemical Process Control	3	
CHE 397—Senior Design II	3	

CHE 499—Professional Development Seminar	0	SAME
Elective outside the major rubric	3	
General Education Core course	3	
Total Hours	14	